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Programme Code: CO5K

Course Name and Code: Seminar and Project Initiation (315003)

A Seminar Report on

Machine Learning In Healthcare

Submitted in partial fulfilment for the award of
THIRD YEAR DIPLOMA IN
COMPUTER ENGINEERING
By

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FOR ACADEMIC YEAR : 2026– 2027

Department of Computer Engineering

S.S.V.P.S's Bapusaheb Shivahirao Deore Polytechnic , Dhule

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(An Affiliated Institute of Maharashtra State Board of Technical Education)

DEPARTMET OF COMPUTER ENGINEERING

Certificate

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Place : Dhule

Date : .....

Signature:

Name : Prof. N. D. Patel

Subject Teacher

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Name : Prof. N. D. Patel

Head of the Department

Seal of Institution

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ABSTRACT

Machine Learning (ML) is a branch of Artificial Intelligence that enables computer systems to learn patterns from data and use those patterns to make predictions or support decisions. Healthcare produces large amounts of information, including electronic health records, laboratory results, medical images, prescriptions and patient-monitoring data. When handled carefully, ML methods can help healthcare professionals identify patterns that may be difficult to detect manually.

This report studies the role of machine learning in healthcare, with emphasis on disease-risk prediction, medical-image analysis, early diagnosis, patient monitoring, personalised care and hospital operations. Common techniques include supervised learning, unsupervised learning, deep learning and natural language processing. The report also discusses data quality, privacy, bias, explainability, clinical validation and responsible deployment. ML should support, not replace, qualified healthcare professionals.

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CHAPTER 1

INTRODUCTION

Healthcare systems increasingly use digital records, diagnostic images, laboratory reports, wearable devices and remote-monitoring tools. These sources contain useful information about patient health, but the volume and complexity of the data can make it difficult to review every detail manually. Machine learning offers methods for learning statistical patterns from historical examples and applying those patterns to new data.

Machine learning is a part of Artificial Intelligence (AI). Instead of relying only on fixed, manually written rules, an ML model is trained using examples. For instance, a model may learn from labelled clinical records to estimate a risk category, or from annotated medical images to identify visual patterns that require further review. Its output is a prediction or score and should be interpreted with clinical context.

In healthcare, ML can support screening, triage, diagnostic assistance, patient-risk estimation, treatment research, drug discovery and hospital resource management. The value of these systems depends not only on algorithm accuracy but also on data quality, fairness, workflow fit, usability and evidence that the system works safely in the intended setting.

Machine learning is needed because clinical data are growing rapidly and timely analysis can help professionals prioritise cases, identify trends and plan resources. However, a prediction is not a confirmed diagnosis. Models should be validated using appropriate data, and their limitations must be communicated clearly to healthcare professionals.

The aim of this report is to explain the basic concepts, workflow, applications, benefits and limitations of machine learning in healthcare, and to identify the resources and safeguards needed for responsible use. This is an educational literature-based report, not a clinically validated diagnostic product.

CHAPTER 2

LITERATURE REVIEW

2.1 AI in Health and Medicine

Rajpurkar, Chen, Banerjee and Topol review advances in AI for health and medicine, including medical image analysis, clinical applications and human–AI collaboration. The review highlights the potential for AI to assist clinicians while identifying challenges such as data quality, bias, and the need for clinical validation.

2.2 Shifting Machine Learning for Healthcare from Development to Deployment

Zhang, Xing, Zou and Wu examine machine learning for healthcare through a data-centric perspective. They emphasise data quality, data availability and the difficulties of moving models from development to real-world deployment. Their review discusses data shifts and methods intended to improve the reliability of healthcare ML.

2.3 A Guide to Deep Learning in Healthcare

Esteva and colleagues provide an overview of deep learning methods and their applications in healthcare. The work describes the potential of neural networks in complex data analysis while highlighting the need for careful evaluation, suitable datasets and collaboration between technical and clinical experts.

2.4 High-Performance Medicine: Human and Artificial Intelligence

Topol discusses how human expertise and artificial intelligence may complement each other in medicine. The article argues for thoughtful integration of AI into clinical work and stresses that the goal should be better care rather than replacing the clinician–patient relationship.

2.5 WHO Guidance on Ethics and Governance of AI for Health

The World Health Organization guidance considers ethical and governance requirements for AI used in health. It identifies principles including protection of autonomy, safety, transparency, accountability, inclusiveness and privacy, and calls for human oversight throughout development and use.

2.6 Machine Learning for Clinical Decision Support

Clinical decision-support research explores how predictions and alerts can be integrated into care workflows. These systems need to be assessed for accuracy, calibration, usability and impact on clinical decisions, not only for technical performance on a test dataset.

A key lesson from clinical ML research is that the dataset should reflect the intended population and environment. Data leakage, missing values and inconsistent measurement can lead to overly optimistic evaluation results or unreliable predictions after deployment.

2.7 Medical Image Analysis Using Deep Learning

Research on deep learning for medical imaging describes automated analysis of X-rays, CT, MRI, retinal images and pathology slides. These tools may highlight areas for closer review, but image quality, equipment differences and population representation affect their reliability.

2.8 Machine Learning for Disease-Risk Prediction

Risk-prediction studies use patient characteristics, history and laboratory measurements to estimate the likelihood of a condition or outcome. A risk score may support follow-up decisions but does not establish a diagnosis, and thresholds must be validated for the intended population.

2.9 Privacy and Security in Healthcare AI

Healthcare ML relies on sensitive information, so privacy and security must be addressed from the beginning. Appropriate controls may include data minimisation, access restrictions, encryption, audit logs and use of approved datasets under institutional policies.

2.10 Fairness, Bias and Explainability

Research on trustworthy AI highlights that models may reproduce bias present in historical data or perform differently across patient groups. Subgroup evaluation, transparent documentation and human oversight help identify limitations and support accountable use.

2.11 Model Drift and Real-World Deployment

A model validated in one setting may perform differently in another hospital or over time as devices, populations and clinical practice change. Monitoring, local validation, incident review and controlled updates are therefore important after deployment.

2.12 Research Gap and Project Direction

Existing research shows that machine learning can support medical image analysis, disease-risk prediction and clinical decision support. However, model performance can be affected by incomplete or unbalanced data, differences between hospitals, privacy requirements and limited explainability. A model that performs well on one dataset may not provide the same results in another clinical setting. Therefore, there is a need to study a practical healthcare ML approach that uses suitable data, careful validation, privacy protection and human supervision. The proposed seminar focuses on understanding these requirements rather than treating an ML prediction as a replacement for professional medical judgement.

2.13 Conclusion of Literature Review

The reviewed literature highlights both the opportunities and limitations of machine learning in healthcare. AI and deep-learning methods may help analyse complex clinical data, while data-centric research stresses the importance of data quality and real-world validation. Ethical guidance further emphasises privacy, fairness, transparency, accountability and human oversight. These findings provide the foundation for this report, which studies healthcare applications of machine learning and the safeguards required for its responsible use.

Table No. 2.1: Literature Review:

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	Review of AI in medicine	Medical AI methods	Published studies	Summarise imaging and clinical AI applications.	Clinical usefulness and safety need validation.
2	Data-centric ML review	ML and data-quality methods	Healthcare datasets in reviewed studies	Explain deployment, data quality and data shift.	Performance may change in new settings.
3	Deep-learning review	Neural networks	Medical images and clinical data	Describe deep learning in healthcare.	Requires suitable data and expert evaluation.
4	Human–AI collaboration	Clinical AI systems	Clinical research literature	Explore how AI can complement clinicians.	Technology must fit clinical workflows.
5	Ethics and governance guidance	Governance principles	Health AI use cases	Promote safe, transparent and accountable use.	Requires implementation and oversight.
6	Clinical decision support	Classification and regression	Clinical records	Support risk estimates and prioritisation.	Poor integration can create alert fatigue.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
7	Clinical data quality	Preprocessing and validation	Electronic health records	Reduce errors and leakage in model building.	Missing and inconsistent data remain common.
8	Medical image analysis	CNNs and deep learning	X-ray, CT, MRI and retinal images	Highlight patterns for professional review.	Equipment and population shifts affect results.
9	Disease-risk prediction	Logistic regression, trees, ensembles	Clinical measurements and history	Estimate risk for a defined outcome.	Risk scores are not diagnoses.
10	Privacy and security	Access control, encryption, minimisation	Sensitive health information	Protect patient data through the ML lifecycle.	Controls must be maintained and audited.
11	Fairness and explainability	Subgroup evaluation and interpretation tools	Representative evaluation datasets	Identify bias and understand model limitations.	Explanations may not prove correctness.
12	Deployment and monitoring	Drift monitoring and version control	Post-deployment performance data	Detect changing performance after launch.	Needs ongoing resources and clear ownership.

CHAPTER 3

INFORMATION

1 . Overview of Machine Learning in Healthcare:

Machine learning is a software technique that learns patterns from examples and uses them to estimate outputs for new data. In healthcare, the inputs may include patient measurements, medical images, text notes or sensor signals. The output should be used within a clearly defined clinical or operational purpose.

2. Types of Machine Learning:

Supervised learning uses labelled examples for classification or regression. Unsupervised learning finds structure in unlabelled data. Deep learning uses multi-layer neural networks for complex inputs such as images and text. Natural language processing analyses written information. Each approach has strengths and limitations and must be selected according to the task.

3. Need for Machine Learning in Healthcare:

Healthcare data are complex and continue to grow. Machine learning may help summarise patterns, prioritise cases, support image review and forecast service demand. Its usefulness depends on high-quality data, appropriate evaluation and integration into professional workflows. Automated predictions should not be treated as a substitute for clinical assessment.

4. Working Principle:

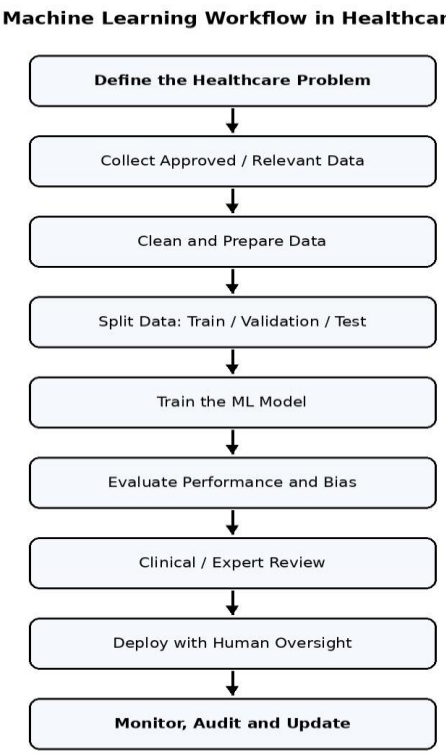


Fig 3.1: Working of Machine Learning in Healthcare

The general working process of a healthcare ML system consists of the following steps:

1. **Define the problem:** Specify the intended task, target population and type of output required.
2. **Collect data:** Use relevant and authorised records, images, measurements or monitoring signals.
3. **Prepare the data:** Check missing values, inconsistent formats, errors and possible data leakage.
4. **Train the model:** Select an appropriate algorithm and learn patterns from the training dataset.
5. **Evaluate the model:** Test it on separate data using metrics suitable for the intended task.
6. **Review the results:** Check limitations, fairness, calibration and performance across relevant groups.
7. **Validate clinically:** Ask domain experts to assess safety, usefulness and workflow fit.
8. **Deploy securely:** Integrate the model with access controls and clear user instructions.
9. **Review predictions:** A qualified professional interprets the output in the clinical context.

10. **Monitor performance:** Track errors, changes in data and potential model drift.

11. **Document changes:** Record versions, limitations, incidents and updates.

12. **End :** Continue use only while the system remains appropriately validated and monitored.

5. System Architecture and Main Components:

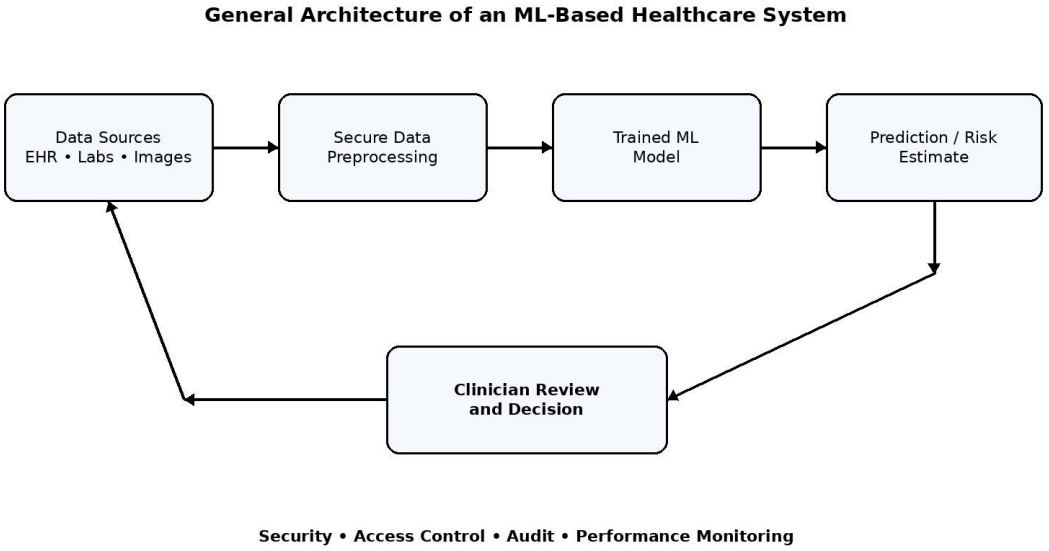


Fig 3.2: Architecture of Machine Learning in Healthcare

The main components may include a data source, secure data-preparation module, trained ML model, output interface and monitoring process. Access controls, encryption, audit logging and clear data-retention rules should be considered from the beginning.

6. Technologies and Techniques Used:

Python is commonly used for educational ML prototypes. pandas and NumPy support data handling, while scikit-learn provides common algorithms and evaluation tools. PyTorch or TensorFlow may be used for deep learning. Tools should be selected according to the task, available resources and approved dataset.

7. Example of a Healthcare ML Workflow:

A hospital may have historical records containing age group, relevant clinical measurements and a confirmed outcome. An authorised research team can prepare the data and train a model to estimate risk for a defined task.

The model's performance should then be measured on separate records not used during training. Sensitivity, specificity, precision, calibration and subgroup results may be considered as appropriate. Any output is a decision-support estimate and must not be presented as a confirmed diagnosis.

8. Responsible Use and Monitoring:

Healthcare data are sensitive and model errors can affect decisions. Systems should be validated for the intended population, reviewed by qualified professionals, protected with appropriate security controls and monitored after deployment. Patients and clinicians should receive clear information about the model's role and limitations.

CHAPTER 4

ADVANTAGES AND DISADVANTAGES

4.1 Advantages

1. Can analyse large datasets and help identify patterns that may be difficult to detect manually.
2. May support earlier risk identification and prioritisation when appropriately validated.
3. Can assist medical-image review by highlighting areas for professional assessment.
4. Can help forecast hospital demand, appointments, bed occupancy and staffing needs.
5. Supports research by finding trends and generating hypotheses for further study.

4.2 Disadvantages

1. Model performance depends on data quality and representativeness.
2. Incorrect predictions can occur, and some models are difficult to explain.
3. Bias in historical data can lead to unequal performance across patient groups.
4. Healthcare data require strong privacy and cybersecurity safeguards.
5. Integration, validation, staff training and ongoing maintenance require resources.
6. Model performance can change when populations, devices or clinical practices change.

CHAPTER 5

CONCLUSION

Machine learning can support healthcare by analysing clinical records, medical images, monitoring signals and operational data. Potential applications include disease-risk prediction, image analysis, early warning, research and hospital planning. When properly evaluated, these systems may help professionals prioritise work and make better-informed decisions.

However, healthcare is a high-impact environment. An inaccurate or biased prediction can affect patient care, and sensitive records require strong protection. Good performance on one dataset does not guarantee safe performance elsewhere. Data quality, privacy, explainability, clinical validation, human oversight and post-deployment monitoring must be treated as core requirements. Machine learning should support, not replace, qualified healthcare professionals.

CHAPTER 6
PROJECT INITIATION

6.1 Description of problem statement :

Healthcare organisations manage large amounts of patient information, but manual review of every record can be time-consuming and may not consistently reveal useful patterns. Machine learning can analyse selected data and produce estimates or flags that assist professionals. The challenge is to ensure that the model is accurate enough for its intended purpose, useful in the workflow and safe for the population where it will be used.

The problem includes data quality, privacy, bias and the risk that model outputs may be misunderstood as definitive diagnoses. A responsible system needs authorised data, appropriate evaluation, clinical review and a process for monitoring performance after deployment. This report studies these requirements and does not claim to have built or clinically validated a medical product.

6.2 Scope and Objectives :

The scope includes basic ML concepts, common algorithms, the model-development workflow, healthcare applications, evaluation metrics and responsible-use challenges. The report is educational and literature-based; it does not provide medical advice or diagnose individual patients. A future prototype would require approved data, supervision and evaluation before any real-world use.

Objectives of the project:

1. To understand the basic concepts and types of machine learning.
2. To identify common algorithms used for healthcare-related prediction and analysis.
3. To study applications such as risk prediction, medical imaging and patient monitoring.
4. To review published research and recognised guidance on AI in healthcare.
5. To understand the importance of data preparation and model evaluation.
6. To identify privacy, fairness, explainability and deployment challenges.

6.3 Stakeholders:

Stakeholders are individuals or groups who use, develop, evaluate, govern or may be affected by a healthcare ML system. Their needs should be considered throughout planning, evaluation and deployment.

1. Patients: Need privacy, fairness, safety and clear communication about the use of their data.
2. Doctors and clinicians: Interpret model outputs and retain responsibility for care decisions.
3. Developers and data scientists: Prepare data, train models, document limitations and Evaluate performance.
4. Hospital IT and security teams: Manage system integration, access, availability and cybersecurity.
5. Hospital administration: Reviews resources, workflow fit, governance and measurable benefit.
6. Project guide and evaluators: Guide the academic work and assess technical accuracy And documentation.

6.4 Platform / Equipment / Resources identification :

1. Platform :

The project can be studied and prototyped on Windows or Linux using Python. Visual Studio Code can be used for development, and a framework such as PyTorch or TensorFlow can load or train the selected detection model. A local environment is suitable for a basic prototype; larger experiments may require access to a GPU-enabled machine.

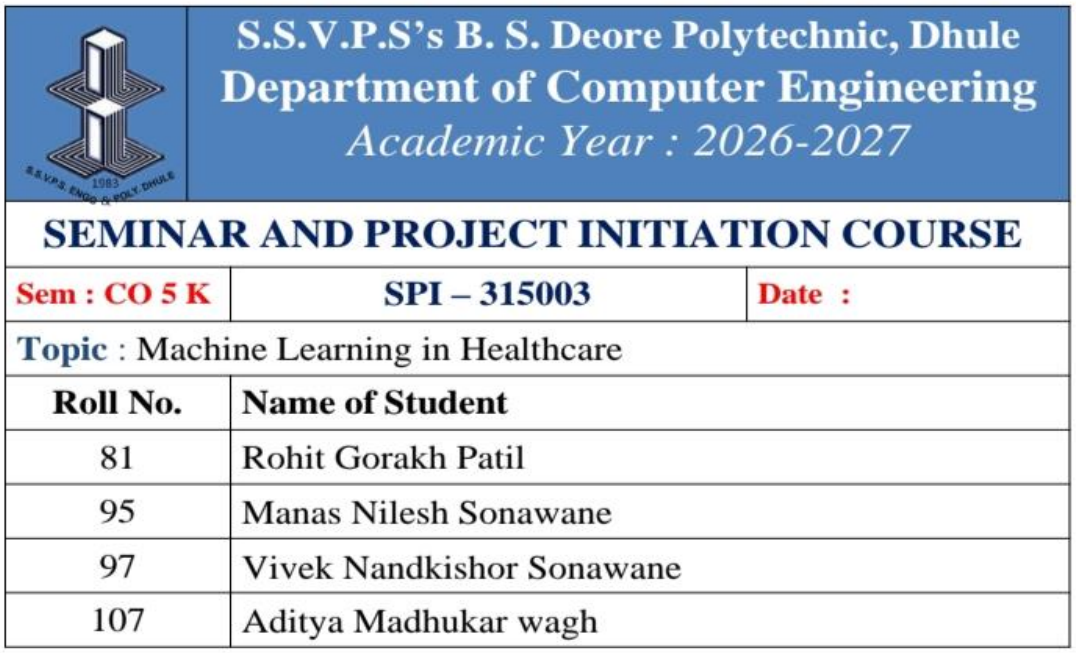
2. Equipment:

- **Computer or laptop:** For development, documentation and testing.
- **Processor:** A modern multi-core processor suitable for Python development.
- **RAM:** 8 GB is a practical starting point for small educational experiments; larger models may require more.
- **Internet connection:** For official documentation and approved package downloads.
- **Storage:** Sufficient space for source code, approved datasets and model files.

3. Resoures:

Resources	Name	Purpose
Operating System	Windows / Linux	Development and testing environment.
Programming Language	Python 3	Primary language for data processing and model implementation.
Code Editor	Visual Studio Code / Jupyter Notebook	Writing, running and debugging code.
Data Handling	pandas and NumPy	Cleaning, transforming and analysing tabular data.
ML Library	scikit-learn	Baseline classification, regression and model evaluation.
Deep Learning (optional)	PyTorch or TensorFlow	Neural networks for image or complexdata experiments.
Visualisation	Matplotlib	Plots for data exploration and evaluation.
Dataset	Approved public or synthetic dataset	Training and testing without unauthorised patient data.

CHAPTER 7
Seminar Presentation



S.S.V.P.S's B. S. Deore Polytechnic, Dhule
Department of Computer Engineering
Academic Year : 2026-2027

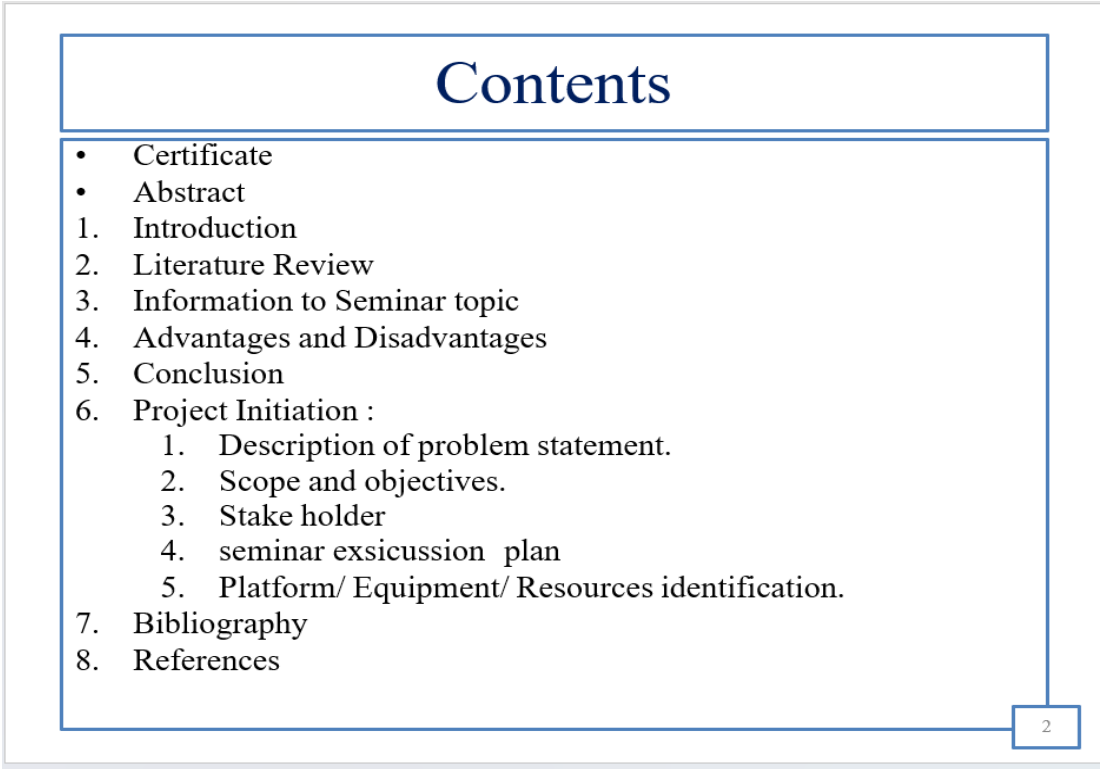
SEMINAR AND PROJECT INITIATION COURSE

Sem : CO 5 K	SPI – 315003	Date :

Topic : Machine Learning in Healthcare

Roll No.	Name of Student
81	Rohit Gorakh Patil
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97	Vivek Nandkishor Sonawane
107	Aditya Madhukar wagh

Fig. No. 7.1: Seminar Title Page



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2. Literature Review
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5. Conclusion
6. Project Initiation :
    1. Description of problem statement.
    2. Scope and objectives.
    3. Stake holder
    4. seminar exsicussion plan
    5. Platform/ Equipment/ Resources identification.
7. Bibliography
8. References

2

Fig. No. 7.2: Content Page

Abstract

- Machine Learning (ML) helps computers learn from data and make predictions. It is used in healthcare to support doctors.
- It helps predict diseases like diabetes and heart disease by analyzing patient data and medical history.
- It can analyze X-rays and MRI scans to help doctors identify diseases at an early stage.
- Machine Learning improves healthcare services, but data privacy and prediction errors are major challenges.

3

Fig. No. 7.3: Abstract Page

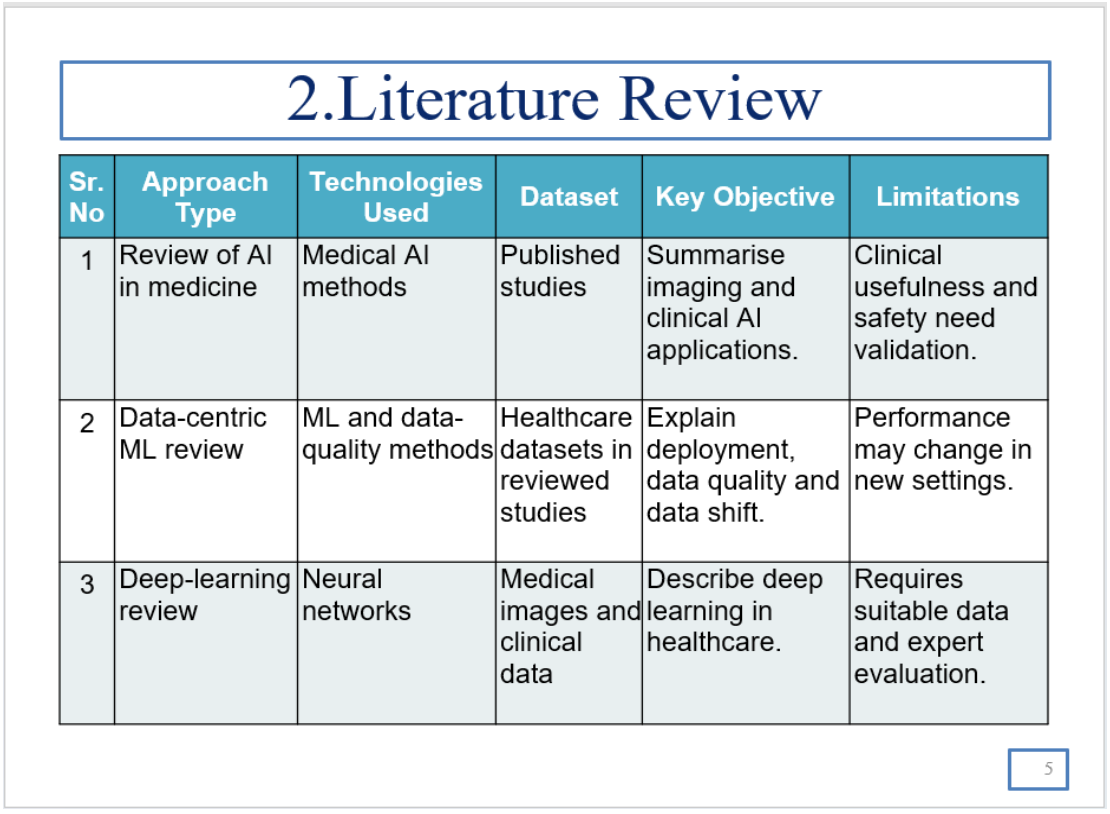
1. Introduction

- Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn from data and make predictions.
- In healthcare, Machine Learning helps doctors predict diseases, analyze medical reports, and identify health problems at an early stage.
- It uses patient information such as age, blood pressure, sugar levels, and medical history to predict health risks
- Machine Learning supports doctors in making decisions and improving the quality of healthcare services.



4

Fig. No. 7.4: Introduction Page

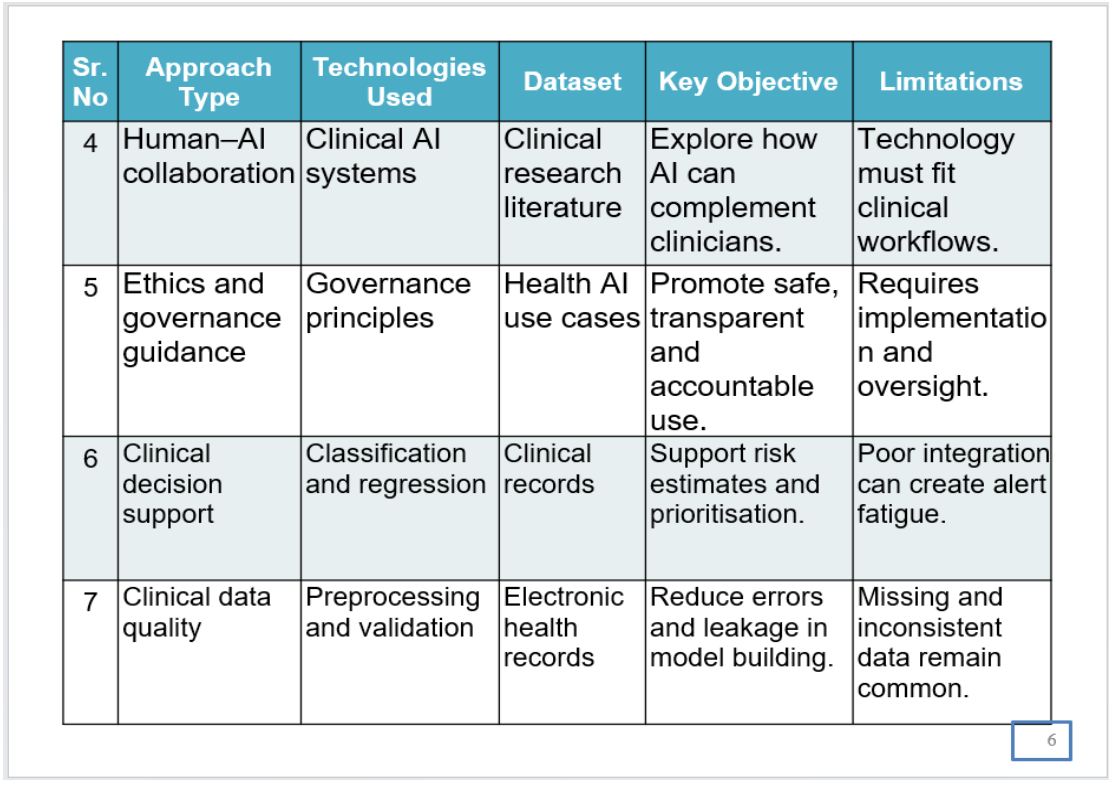


2.Literature Review

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2	Data-centric ML review	ML and data-quality methods	Healthcare datasets in reviewed studies	Explain deployment, data quality and data shift.	Performance may change in new settings.
3	Deep-learning review	Neural networks	Medical images and clinical data	Describe deep learning in healthcare.	Requires suitable data and expert evaluation.

5

Fig. No. 7.5: Literature Review



Sr. No	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
4	Human–AI collaboration	Clinical AI systems	Clinical research literature	Explore how AI can complement clinicians.	Technology must fit clinical workflows.
5	Ethics and governance guidance	Governance principles	Health AI use cases	Promote safe, transparent and accountable use.	Requires implementation and oversight.
6	Clinical decision support	Classification and regression	Clinical records	Support risk estimates and prioritisation.	Poor integration can create alert fatigue.
7	Clinical data quality	Preprocessing and validation	Electronic health records	Reduce errors and leakage in model building.	Missing and inconsistent data remain common.

6

Fig. No. 7.6: Literature Review

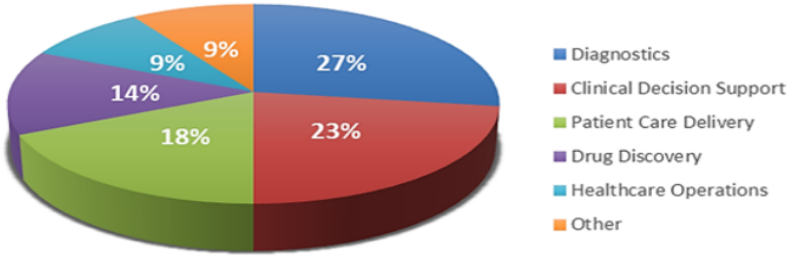
Sr. No	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
8	Medical image analysis	CNNs and deep learning	X-ray, CT, MRI and retinal images	Highlight patterns for professional review.	Equipment and population shifts affect results.
9	Disease-risk prediction	Logistic regression, trees, ensembles	Clinical measurements and history	Estimate risk for a defined outcome.	Risk scores are not diagnoses.
10	Privacy and security	Access control, encryption, minimisation	Sensitive health information	Protect patient data through the ML lifecycle.	Controls must be maintained and audited.
11	Fairness and explainability	Subgroup evaluation and interpretation tools	Representative evaluation datasets	Identify bias and understand model limitations.	Explanations may not prove correctness.
12	Deployment and monitoring	Drift monitoring and version control	Post-deployment performance data	Detect changing performance after launch.	Needs ongoing resources and clear ownership.

7

Fig. No. 7.7: Literature Review

2. Literature Review

- In 2022, Angela Zhang and other researchers studied how Machine Learning can be used in healthcare. They explained the importance of medical data and the challenges of using ML in real hospitals.
- In 2022, Pranav Rajpurkar and other researchers studied AI in healthcare. They explained how AI helps analyze medical images, support doctors, and improve disease diagnosis.
- Major Appllication of ML in Healthcare :-



8

Fig. No. 7.8: Literature Review

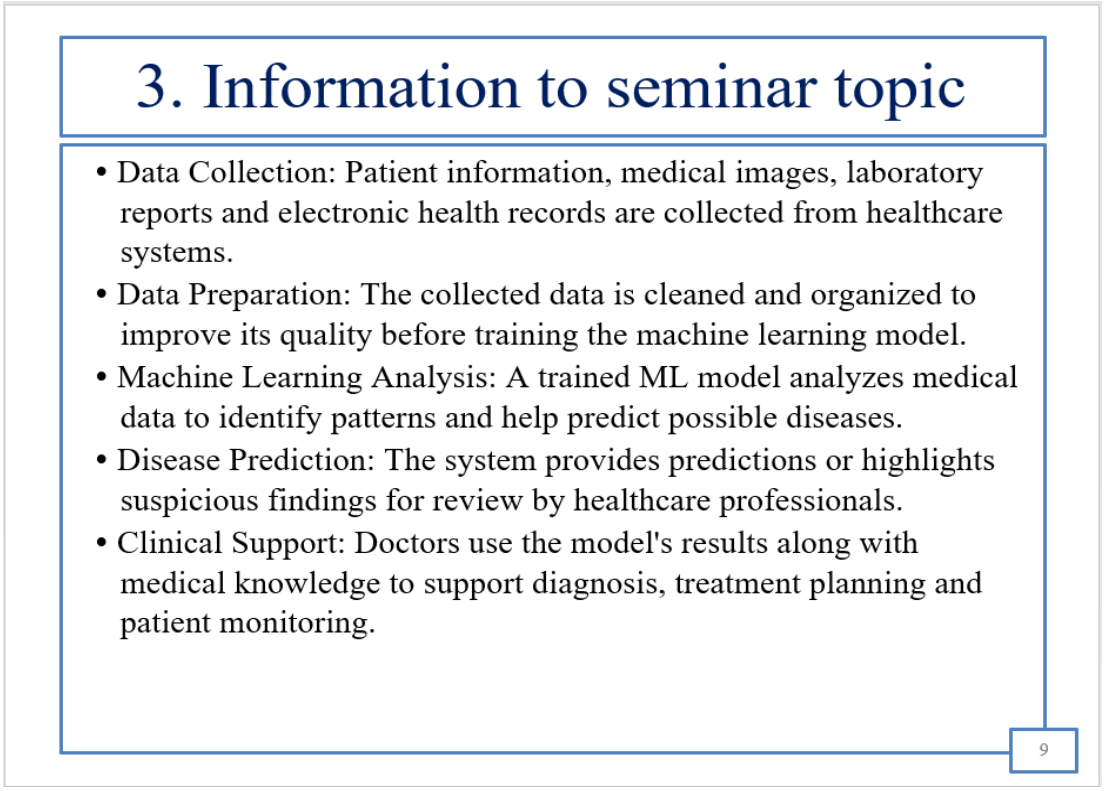


Fig. No. 7.9: Information To Seminar Topic

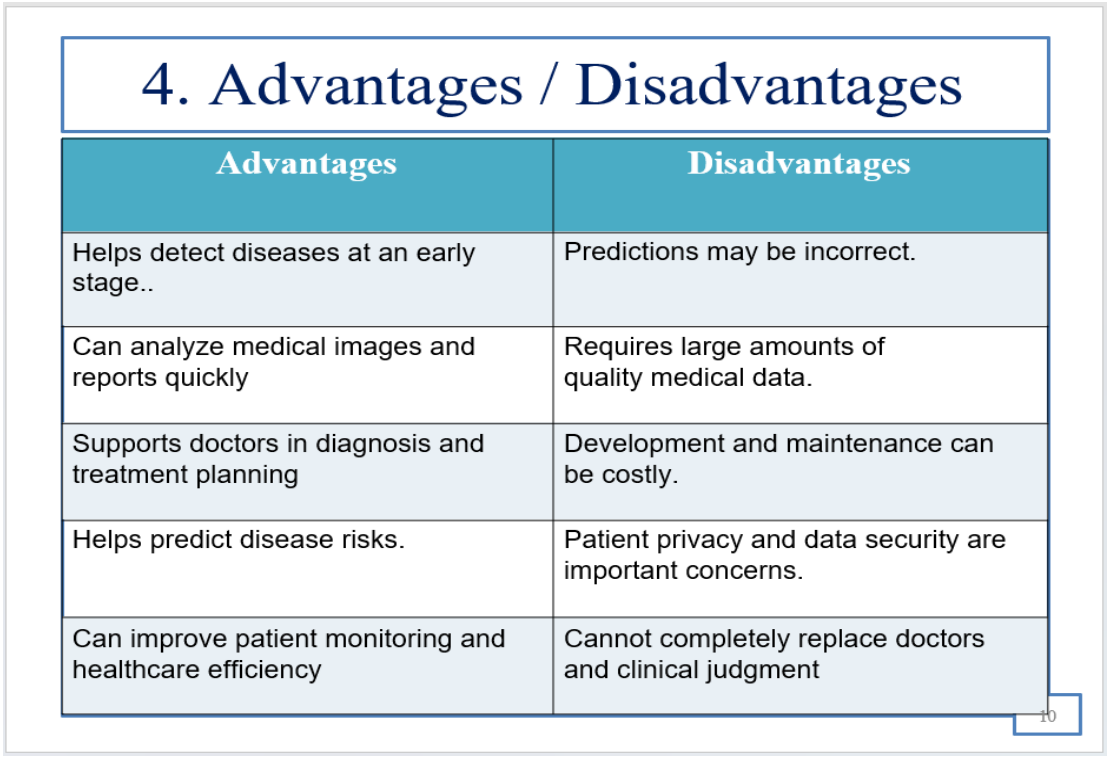


Fig. No. 7.10: Advantage And Disadvantage

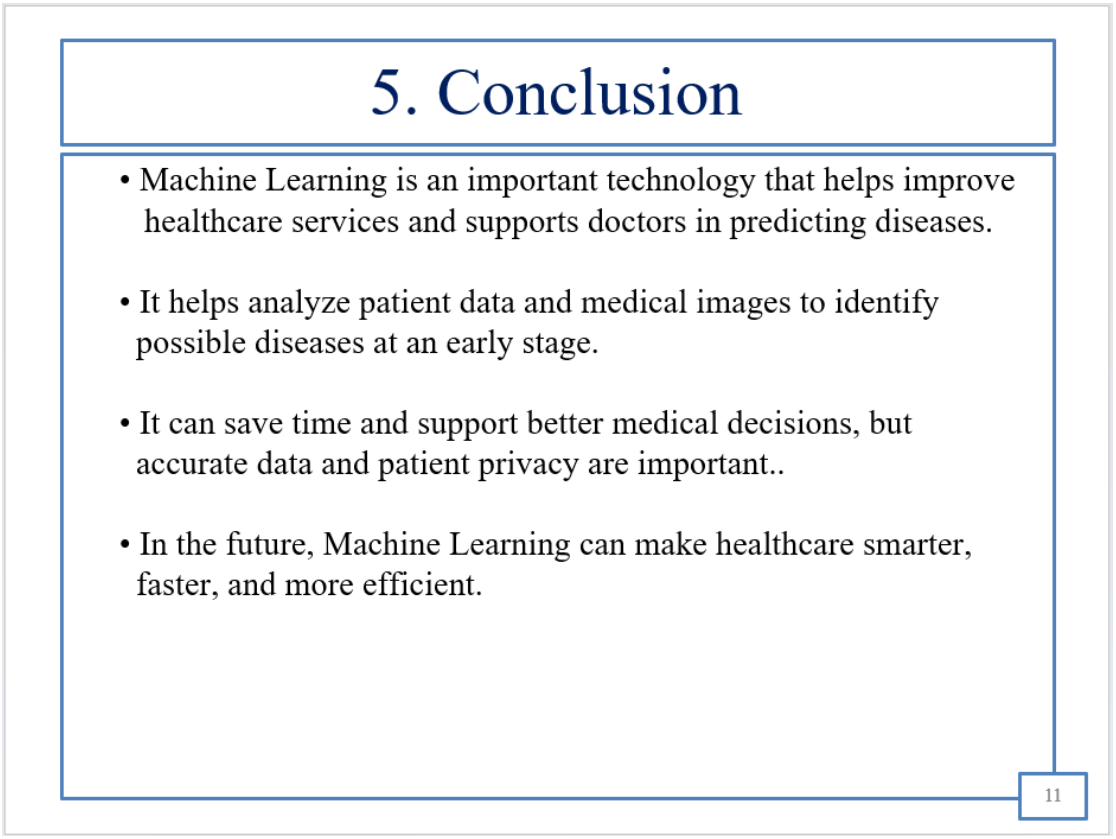


Fig. No. 7.11: Conclusion

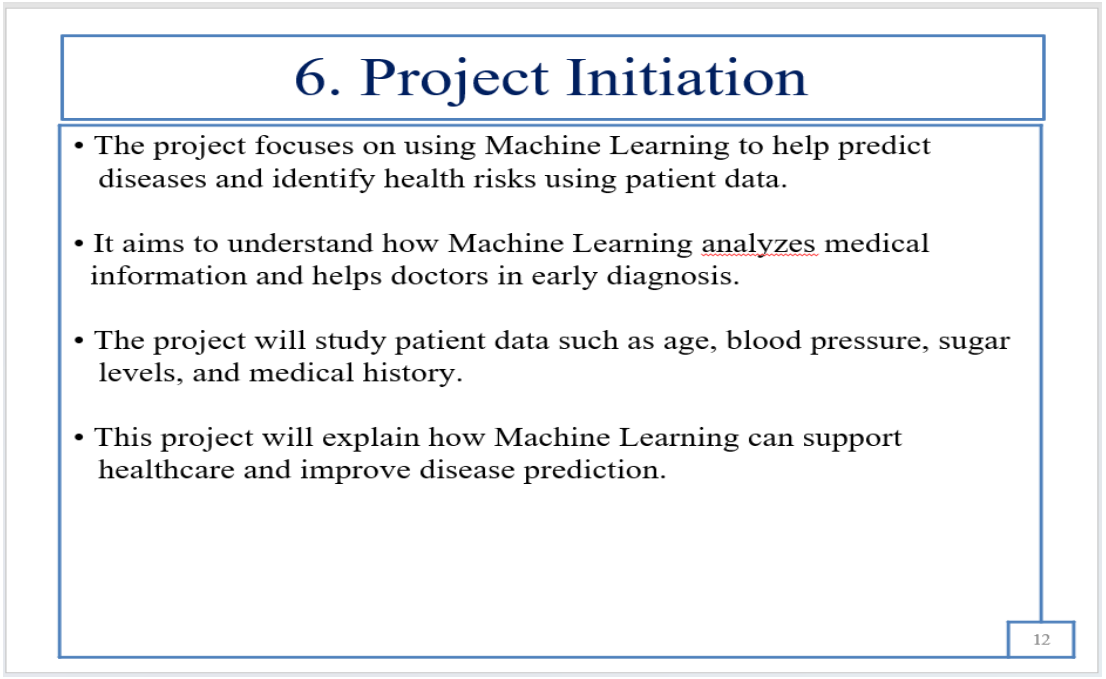


Fig. No. 7.12: Project Initiation

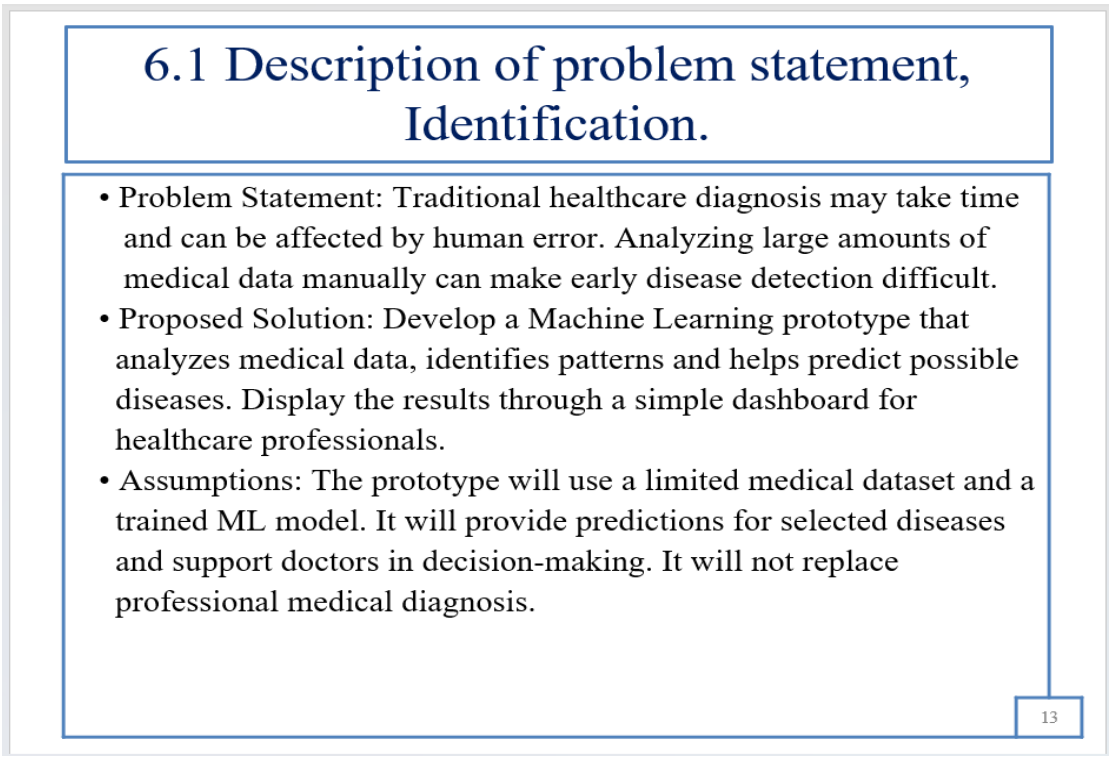


Fig. No. 7.13: Description of problem statement, identification.

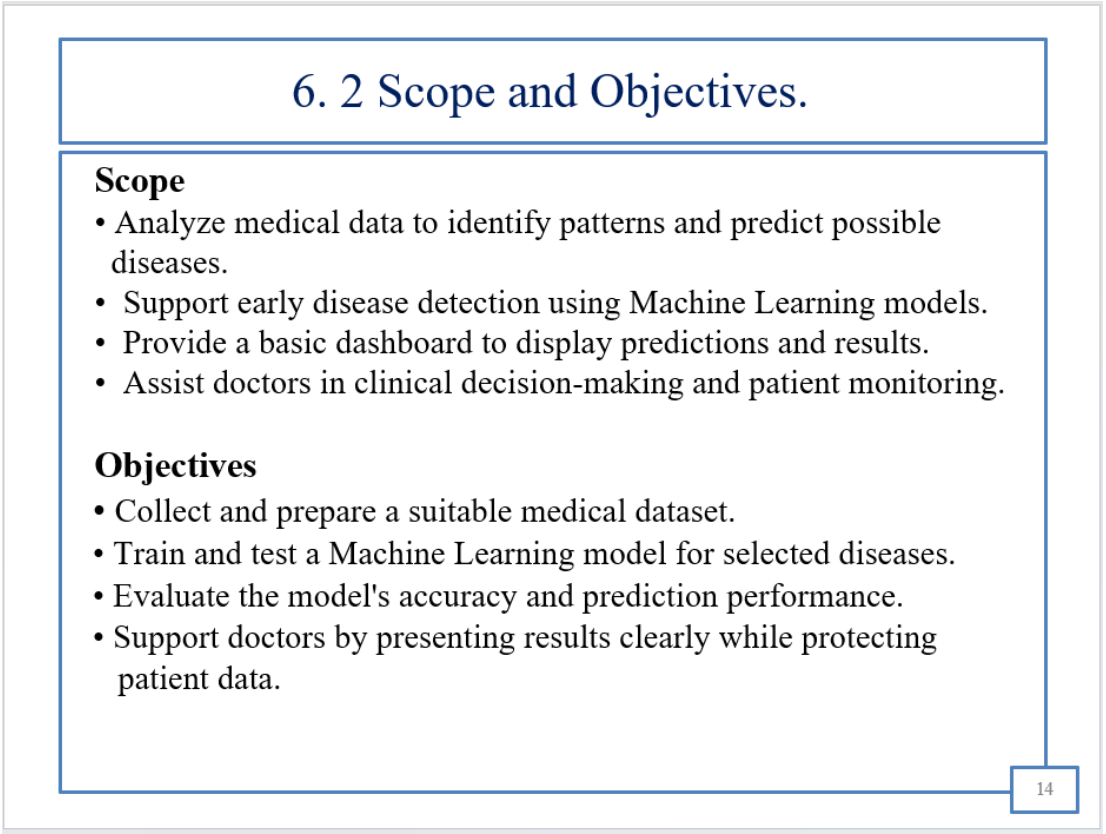


Fig. No. 7.14: Scope And Objectives

6. 3 Stake Holder

Stakeholder	Role/Interest
Doctors	Use ML predictions to support diagnosis and treatment
Patients	Benefit from early disease detection and better monitoring.
Project Team	Designs, develops, tests and documents the ML prototype.
Faculty Guide	Reviews progress and provides technical guidance.
Hospitals	Provide suitable medical data and healthcare requirements.
Data Scientist	Prepare datasets, train models and aluate accuracy.

Fig. No. 7.15: Stake Holder

6. 4 Seminar Exsicussion Plan

Session	Activity	Expected output
1.	Define the problem and project Goals.	Approved problem statement.
2.	Study ML applications in healthcare.	Literature review notes.
3.	Prepare system architecture and workflow.	Architecture diagram
4.	Identify tools, datasets and resources.	Resource list.
5.	Plan model development and testing.	Test Plan and Evaluation criteria.
6.	Review findings and prepare presentation.	Final seminar presentation

Fig. No. 7.16: Seminar Exsicussion Plan

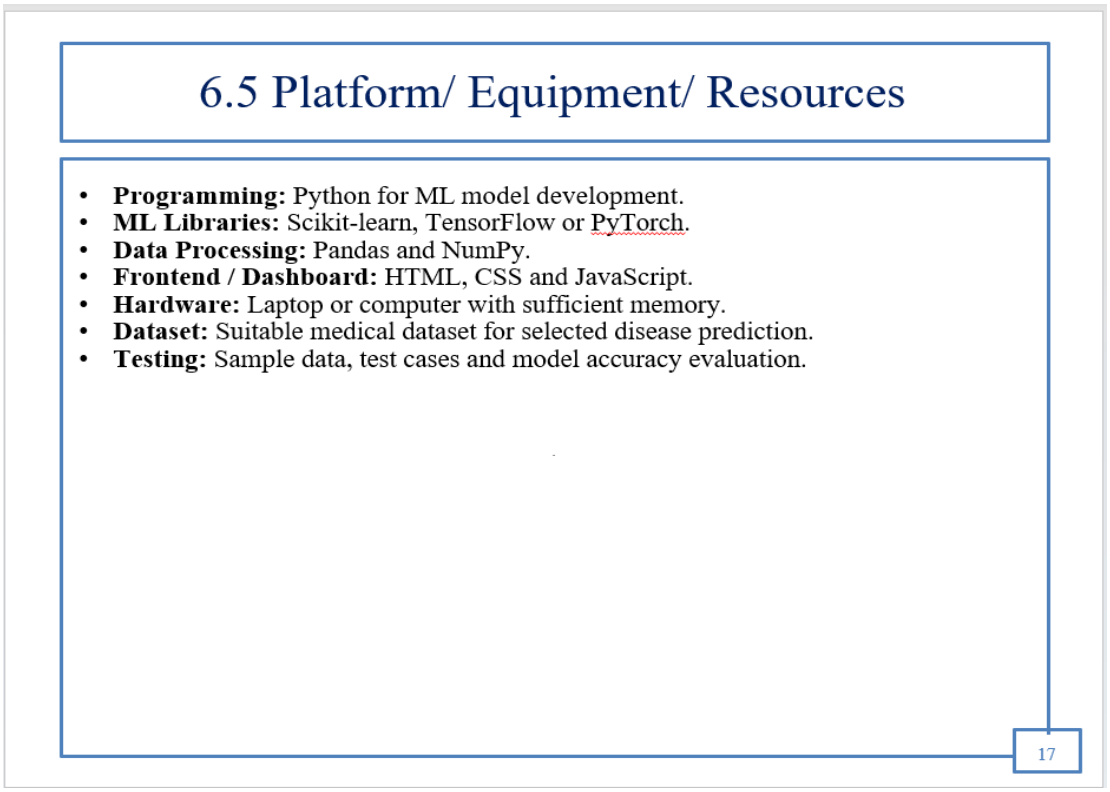


Fig. No. 7.17: Platform/ Equipment/ Resources

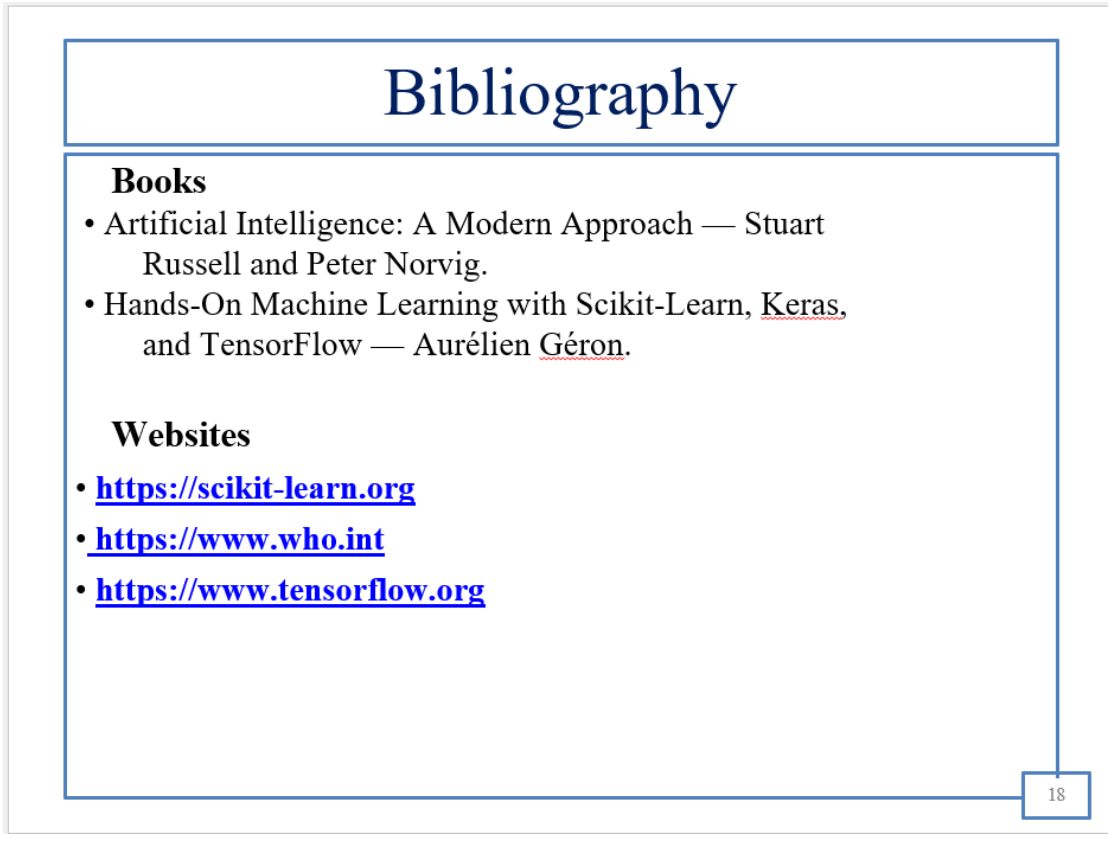


Fig. No. 7.18: Bibliography

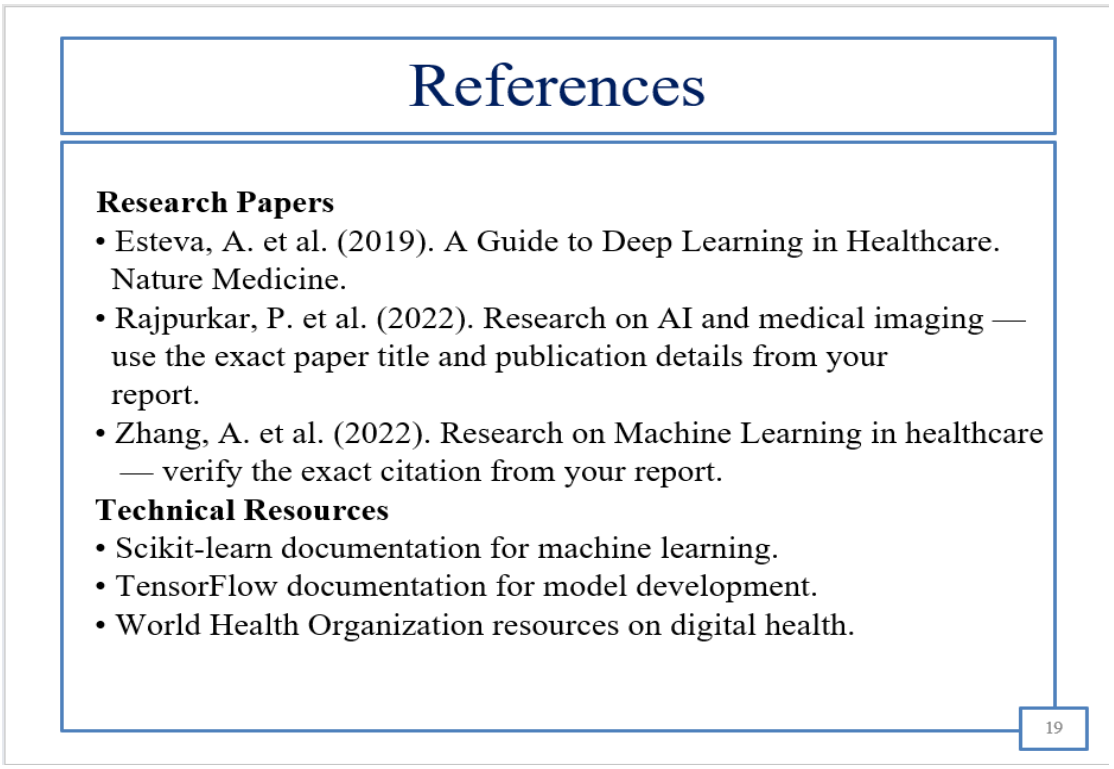


Fig. No. 7.19: References



Fig. No. 7.20: Thank You Slide

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SEMINAR and PROJECT INITIATION Assessment / Evaluation

Roll No	Seat No	Full Name of Students	Enrollment No
81	388568	Rohit Gorakh Patil	24610960266
95	388582	Manas Nilesh Sonawane	24610960282
97	388584	Vivek Nandkishor Sonawane	24610960284
107	388594	Aditya Madhukar Wagh	24610960294

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388568	388582	388584	388594
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign :

Name : Prof. N. D. Patel

(Course Expert/s)